



Finding a length in composite shapes

Task A: Finding lengths from areas of circles and sectors

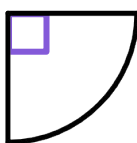
1) Complete the table to find the area of the full circle that the sector comes from.

area of sector	$10\pi \text{ cm}^2$	25.1 cm^2	$31\pi \text{ cm}^2$	$60\pi \text{ cm}^2$
calculation	$10\pi \times 4$	25.1×2	$31\pi \times 4$	$60\pi \div 3 \times 4$
area of full circle	$40\pi \text{ cm}^2$	50.2 cm^2	$124\pi \text{ cm}^2$	$80\pi \text{ cm}^2$

2) The area of this quarter-circle is $900\pi \text{ cm}^2$.

a) Find the area of the full circle that the sector comes from.

$$\text{area of full circle} = 3600\pi$$



b) Using your answer to part a, complete the method to calculate the radius of this quarter-circle.

$$\text{area of full circle} = \pi \times r^2$$

$$3600\pi = \pi \times r^2$$

$$3600 = r^2$$

$$60 \text{ cm} = r$$

3) The area of a semicircle is 39.3 cm^2 .

Complete the method to calculate the diameter of this semicircle.

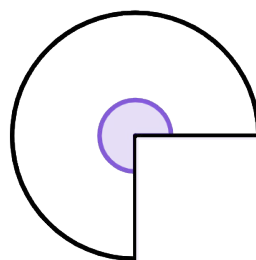
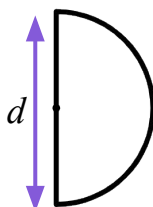
$$\text{area of full circle} = \pi \times r^2$$

$$78.6 = \pi \times r^2$$

$$25.01... = r^2$$

$$5.001... = r$$

$$10 \text{ cm} \approx d$$



4) The area of this three-quarter-circle is 1357.2 cm^2 .

What is the radius of the sector?

$$\text{area of quarter-circle} = 452.4$$

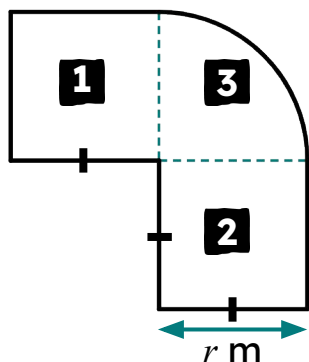
$$\text{area of full circle} = 1809.6$$

$$r = 24 \text{ cm (2 s.f.)}$$



Task B: Finding lengths of composite shapes using algebra

1) This composite shape has an area of 470.7 m^2 .



a) Write down the areas, in terms of r , for the three component parts of this shape.

area of **1** r^2 area of **2** r^2 area of **3** $\frac{1}{4}\pi \times r^2$

b) Show by factorising that the area of the composite shape can be written as:

$$\frac{1}{4}\pi \times r^2 + 2r^2 = r^2 \left(\frac{1}{4}\pi + 2 \right) = r^2 \times 2.785\dots$$

c) Complete the calculation to find the radius of the quarter-circle, given to the nearest integer.

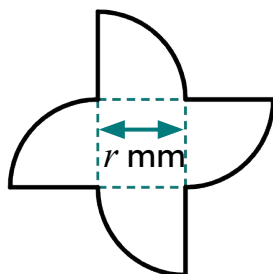
$$r^2 \times 2.785\dots = 470.7$$

$$r^2 = 169$$

$$r = 13 \text{ m}$$

2) This composite shape has an area of $11\,633.7 \text{ mm}^2$.

Complete the steps to calculate the value of r , giving your answer to the closest integer.



area: $\boxed{\pi \times r^2}$ + $\boxed{r^2}$ = $11\,633.7$
circle square

factorise: $r^2 \boxed{(\pi + 1)}$ = $11\,633.7$

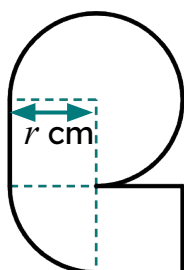
convert π expression into decimal: $r^2 \boxed{(4.14\dots)}$ = $11\,633.7$

divide: r^2 = $\boxed{2808.99\dots}$

square root: r $\approx$ $\boxed{53 \text{ mm}}$

3) This composite shape has an area of 4031 cm^2 .

Find the value of r , giving your answer to the closest integer.



$$2r^2 + \pi r^2 = 4031$$

$$r = 28 \text{ cm (0 d.p.)}$$

